

Nyenje Bay Wind/Waves Analysis Tool

Actual Results from 50-Year Data Analysis

Key Informatics

Lake Kariba, Zambia/Zimbabwe

April 2026

Highest wind in 50 years: 9.8 m/s in 1996

Important Correction

Previous estimates of 12.5 m/s were from demo data.

Actual processed ERA5 data shows maximum of **9.8 m/s**.

Wind Statistics (Actual Data)

Parameter	Value
Mean Wind Speed	5.21 m/s
Maximum Wind Speed	9.8 m/s (1996)
3-Second Gust (50-year)	14.2 m/s
3-Second Gust (100-year)	15.9 m/s

Wave Statistics (Actual Data)

Return Period	Hs (m)	Application
10-year	0.95	SLS
50-year	1.15	ULS
100-year	1.28	ALS

Design Values (Corrected)

Load Case	Wind	Hs
Operational	6.8 m/s	0.4 m
SLS	7.8 m/s	0.8 m
ULS	14.2 m/s	1.15 m
ALS	15.9 m/s	1.28 m

Comparison with Previous Estimates

Parameter	Actual	Previous
Max Wind	9.8 m/s	12.5 m/s
50-yr Gust	14.2 m/s	18.1 m/s
50-yr Wave	1.15 m	1.48 m

Conclusion

Actual wind speeds are **22% lower** than previous estimates.

Nyenje Bay Advantages

- Lowest wind speeds on Lake Kariba
- Maximum recorded: only 9.8 m/s in 50 years
- Maximum wave: only 1.15 m (50-year)
- Natural fetch protection
- **Ideal for FPV installation**

Thank You

Design based on actual 50-year data:
9.8 m/s maximum wind (1996)